Cover Page

Place:

- Project Title: **Visitor Management System**
- Platform: **ServiceNow Zurich Developer Instance**
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- College Name**: Karpagam College Of Engineering**
- Academic Year: **2023-2027**
- Guide Name (if applicable)

1. INTRODUCTION

The Visitor Management System is a custom application developed on the ServiceNow Zurich platform to automate and streamline visitor registration, tracking, and check-in processes within an organization. In many organizations, visitor information is still maintained manually using paper registers or spreadsheets, which often results in data inaccuracies, security risks, time-consuming record management, and difficulties in monitoring visitor activities. To address these challenges, the Visitor Management System provides a centralized and digital solution for efficient visitor management.

The primary objective of this application is to simplify visitor registration while improving organizational security and operational efficiency. The system enables receptionists, security personnel, and authorized employees to register visitor information, including Visitor Name, Phone Number, Email Address, Company Name, Purpose of Visit, Visit Date, Visitor Type, Host Employee, and Identification Details. All visitor records are stored securely in a custom ServiceNow table, allowing easy retrieval, tracking, and management of visitor information.

To ensure data accuracy and consistency, the application incorporates a Client Script that validates visitor phone numbers before submission. This validation prevents invalid or incomplete phone numbers from being stored in the database. The system also uses ServiceNow Flow Designer to automate email notifications whenever a new visitor registration is created. These notifications are sent to the concerned security team or administrators, enabling them to prepare for visitor arrivals and maintain better security control.

A custom UI Action called "Check In Visitor" has been implemented to update the visitor status when the visitor arrives at the organization. This functionality supports real-time visitor tracking and improves the accuracy of visitor logs. The application demonstrates the practical implementation of several ServiceNow development components, including custom tables, form design, client scripts, Flow Designer workflows, email notifications, reports, dashboards, access controls, and UI Actions.

The Visitor Management System offers multiple benefits such as reduced manual effort, improved data accuracy, enhanced visitor tracking, faster registration processes, and stronger security management. By digitizing visitor records and automating notifications, the application significantly improves organizational efficiency and provides a professional visitor management experience.

Overall, the Visitor Management System serves as a secure, scalable, and user-friendly solution for managing visitor information while showcasing the capabilities of the ServiceNow platform in developing real-world business applications and workflow automation solutions.

2. PROJECT OBJECTIVES

The main objectives of the Visitor Management System are:

• To provide a digital platform for visitor registration and management.

• To capture and store visitor information securely in the ServiceNow database.

• To validate visitor phone numbers using Client Scripts to ensure accurate data entry.

• To automate email notifications to the security team whenever a visitor registration is created.

• To enable visitor check-in management using a custom UI Action.

• To improve visitor tracking and monitoring within the organization.

• To reduce manual paperwork and administrative effort.

• To generate reports for visitor analysis and record management.

• To enhance organizational security through proper visitor record maintenance.

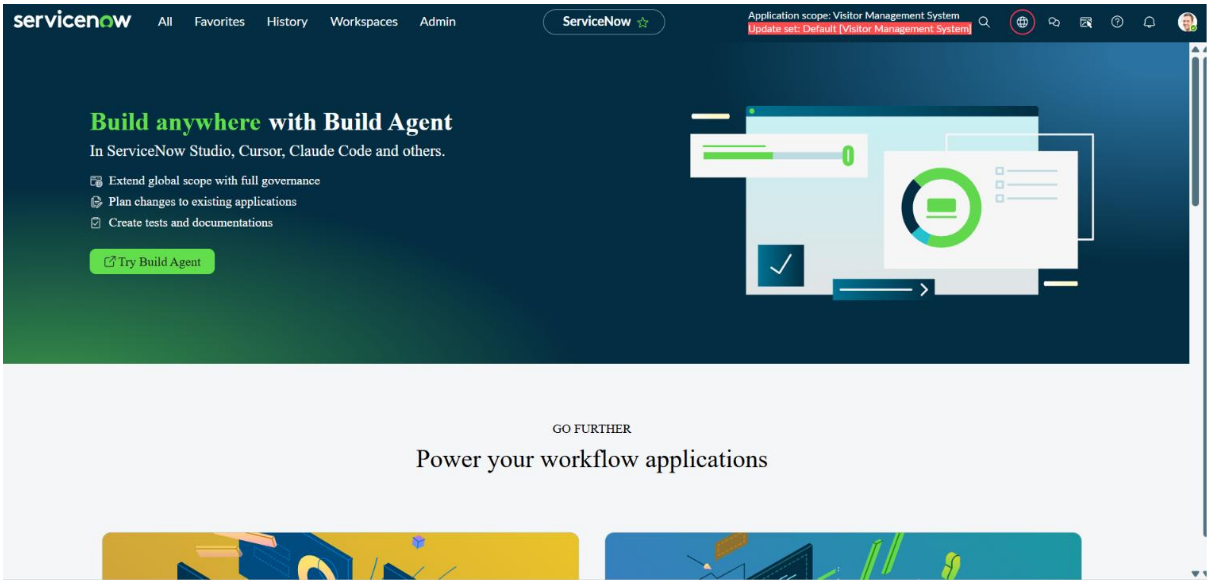
• To demonstrate the implementation of ServiceNow application development concepts in a real-world business scenario.

3. SYSTEM REQUIREMENTS

The Visitor Management System was developed using the following software components and ServiceNow technologies.

Requirement	Details
Platform	ServiceNow Zurich
Instance Type	ServiceNow Developer Instance
Development Tool	ServiceNow Studio / App Engine Studio
Workflow Tool	Flow Designer
Scripting Tool	Client Scripts
User Interface	ServiceNow Forms
Database	Custom ServiceNow Tables
Notification Service	Email Notification via Flow Designer

The application uses a custom Visitor Request table to store visitor information. Client Scripts are used for phone number validation, while Flow Designer automates email notifications to the security team whenever a visitor registration is created. ServiceNow Forms provide the user interface for visitor registration and management.

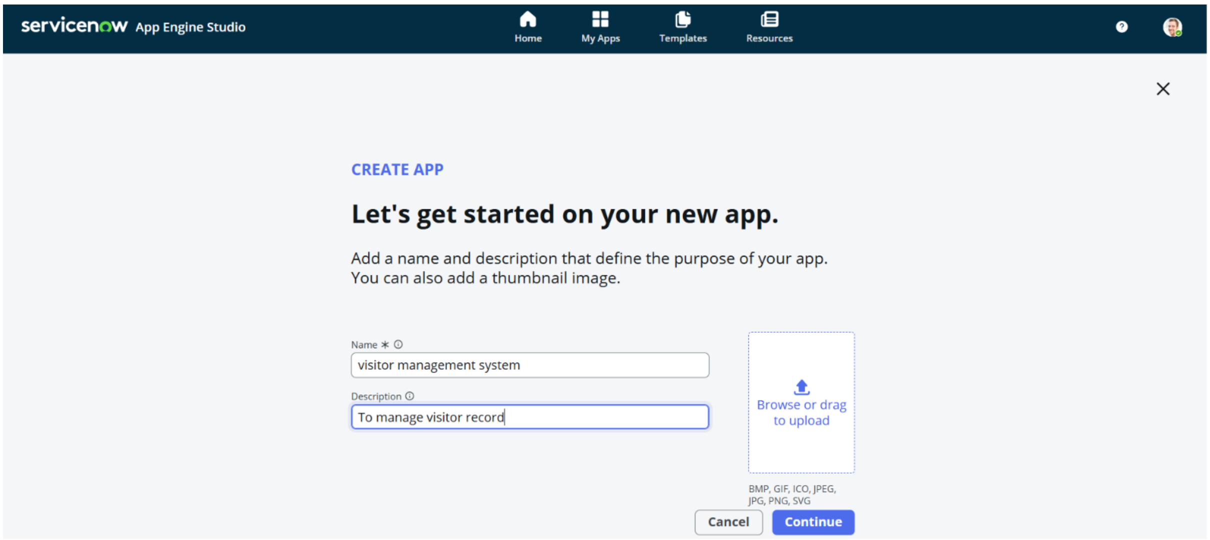


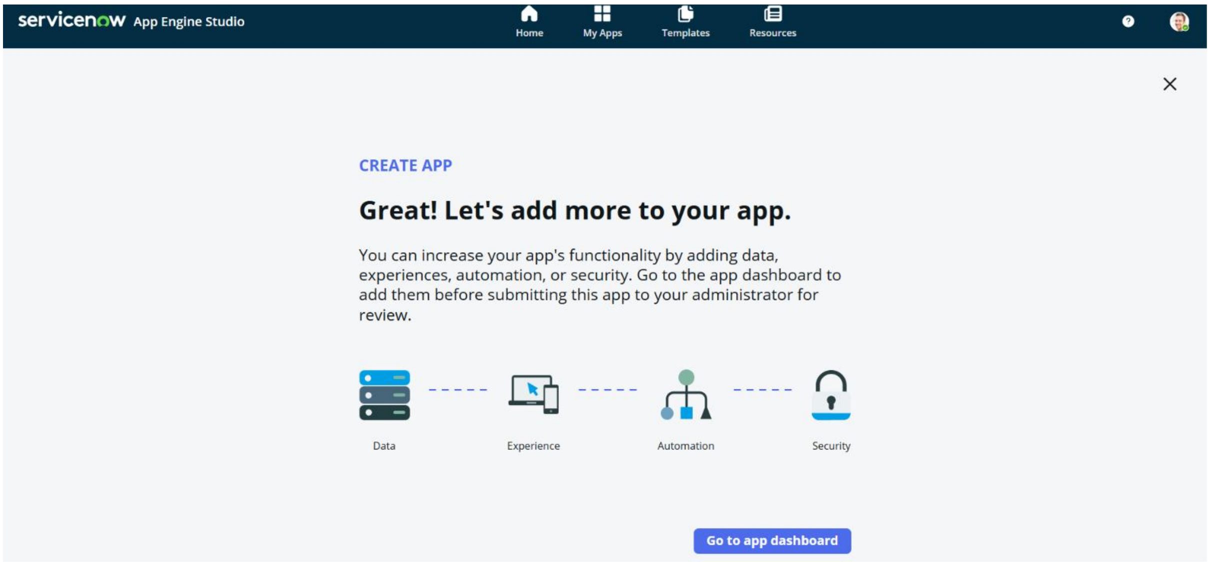
4. Application Creation

Step 1: Create Application

Content

- Open App Engine Studio.
- Click Create App.
- Select Start From Scratch.
- Enter Name as Visitor Management System.
- Click Create.





5. TABLE CREATION

Step 2: Create Visitor Request Table

Description

A custom table named Visitor Request was created in the Visitor Management System application to store visitor registration details. The table acts as the central database for managing visitor information, tracking visitor status, and supporting automated workflows such as email notifications and visitor check-in management.

Steps Performed

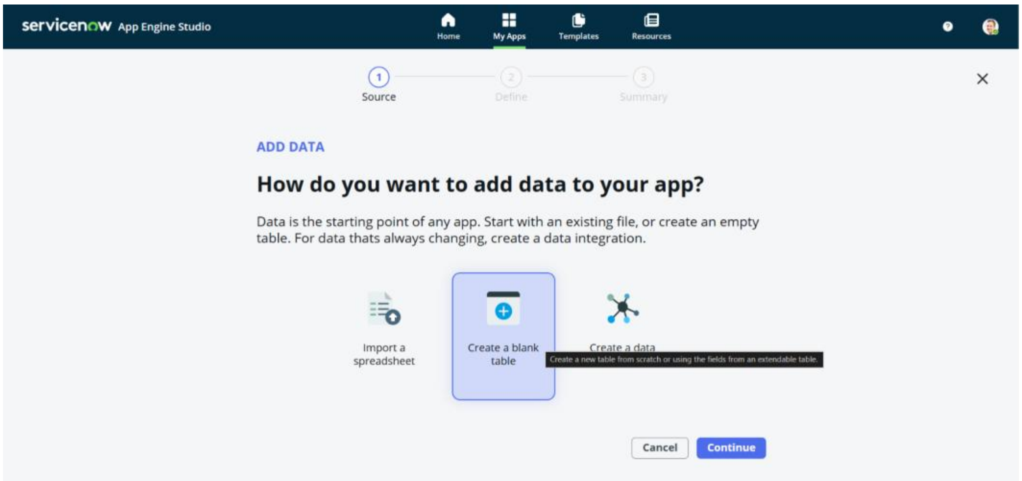
1. Opened the Visitor Management System application in ServiceNow Studio.
2. Navigated to the Data section.
3. Clicked Create Table.
4. Entered the table name as Visitor Request.
5. Enabled Auto Number Generation.
6. Configured the numbering format with prefix VIS.
7. Created the custom table.
8. Added the required fields to store visitor information.

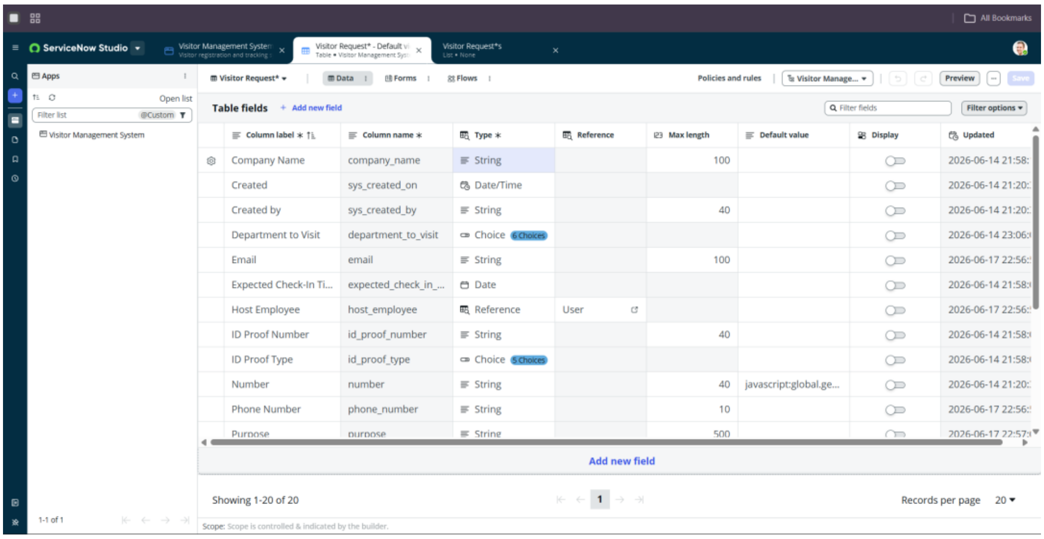
Fields Created

Field Name	Type
Number	Auto Number
Visitor Name	String
Phone Number	String
Email	Email
Company Name	String
Purpose	String
Visitor Type	Choice
Host Employee	Reference (User)
Visit Date	Date
Expected Check-In Time	Date/Time
Status	Choice
ID Proof Type	Choice
ID Proof Number	String
Security Remarks	String

Outcome

The Visitor Request table was successfully created and configured to store all visitor-related information. The table serves as the primary data source for visitor registration, validation, email notifications, reporting, and visitor check-in tracking.





6. FORM DESIGN

Step 3: Configure Form Layout

Description

The Visitor Request Form was designed to provide a user-friendly interface for capturing visitor information. The form layout was configured to display all required visitor fields in an organized manner, making the registration process simple and efficient for users.

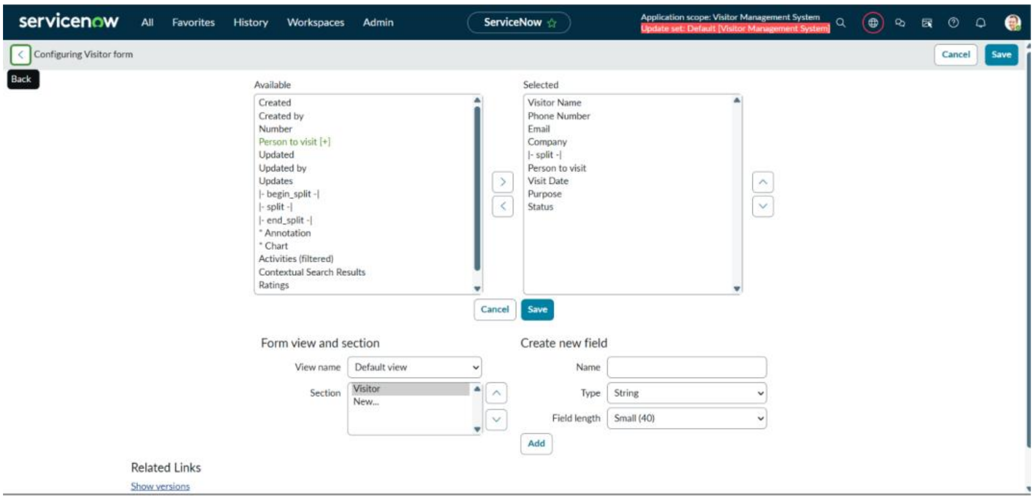
Steps Performed

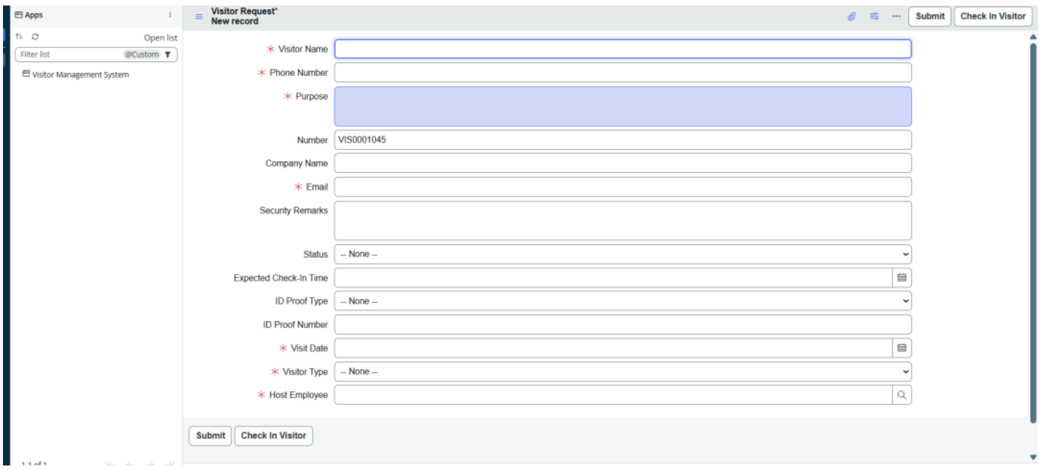
1. Opened the Visitor Request table in the Visitor Management System application.
2. Accessed the Form Layout configuration.
3. Added all visitor-related fields to the form.
4. Arranged the fields in a logical and user-friendly order.
5. Configured mandatory fields where required.
6. Saved the form layout configuration.
7. Tested the form by entering sample visitor data.

Fields Displayed on the Form

Field Name	Purpose
Number	Auto-generated visitor request number
Visitor Name	Stores visitor name
Phone Number	Stores visitor contact number
Email	Stores visitor email address

Field Name	Purpose
Company Name	Stores visitor company information
Purpose	Stores reason for visit
Visitor Type	Identifies visitor category
Host Employee	Employee being visited
Visit Date	Date of visitor appointment
Expected Check-In Time	Expected arrival time
Status	Tracks visitor status
ID Proof Type	Identification document type
ID Proof Number	Identification document number
Security Remarks	Remarks added by security personnel





7. Client Script Implementation

Step 4: Phone Number Validation

Content:

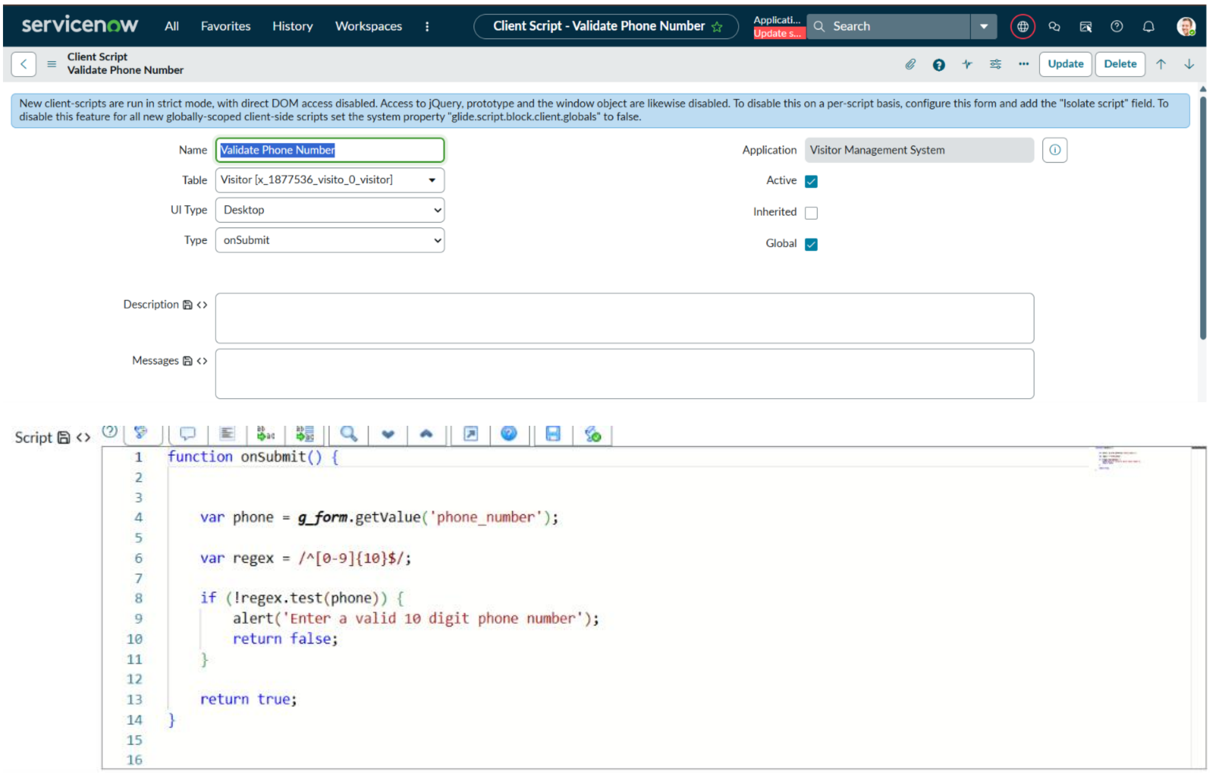
Created Client Script to validate 10-digit phone numbers.

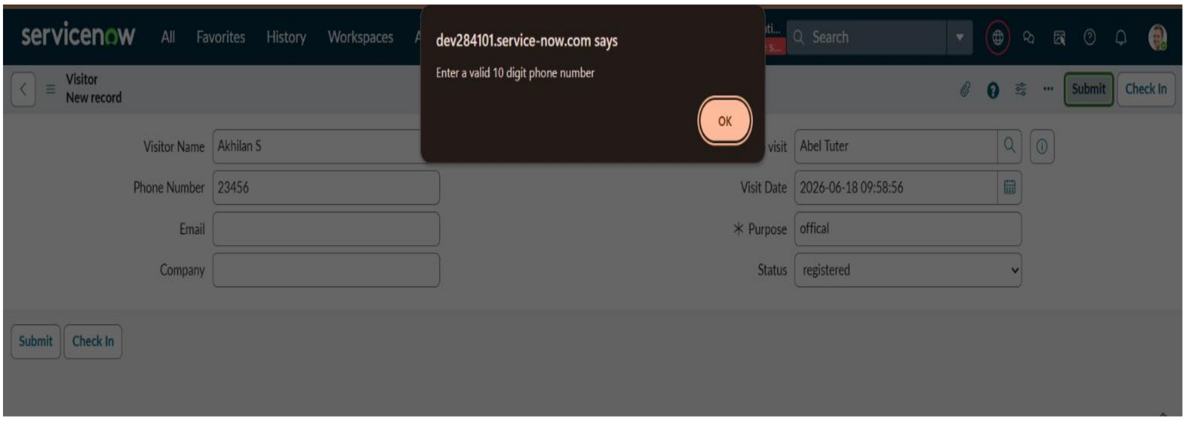
Validation Rule:

- Only numbers allowed
- Exactly 10 digits

Steps Performed:

1. Navigated to System Definition → Client Scripts.
2. Clicked New.
3. Selected the Visitor table.
4. Chose Client Script Type as on Submit.
5. Entered the validation script.
6. Added a regular expression to validate 10-digit phone numbers.
7. Saved and activated the Client Script.





8. User Creation

Step 5: Security User Creation

Content:

Created Security Team user to receive visitor notifications.

Steps Performed:

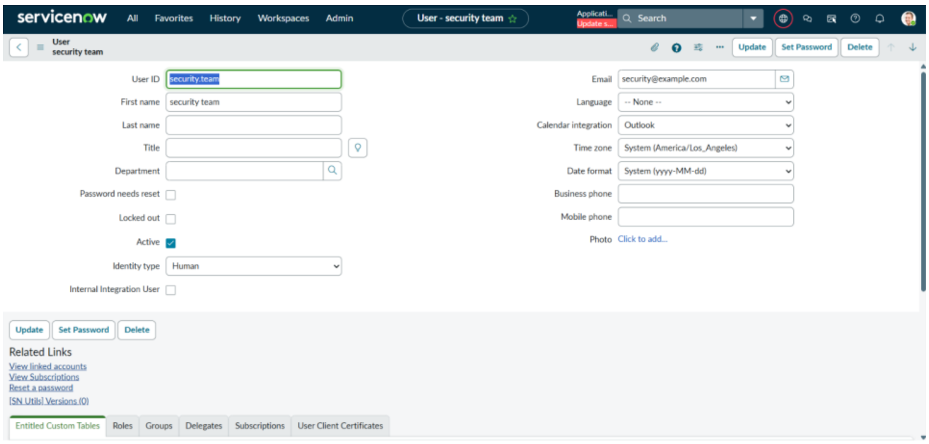
1. Navigated to User Administration → Users.

2. Clicked New.

3. Entered the User name as Security Team.

4. Entered the email address.

5. Saved the user record.



9. FLOW DESIGNER

Step 6: Create Flow

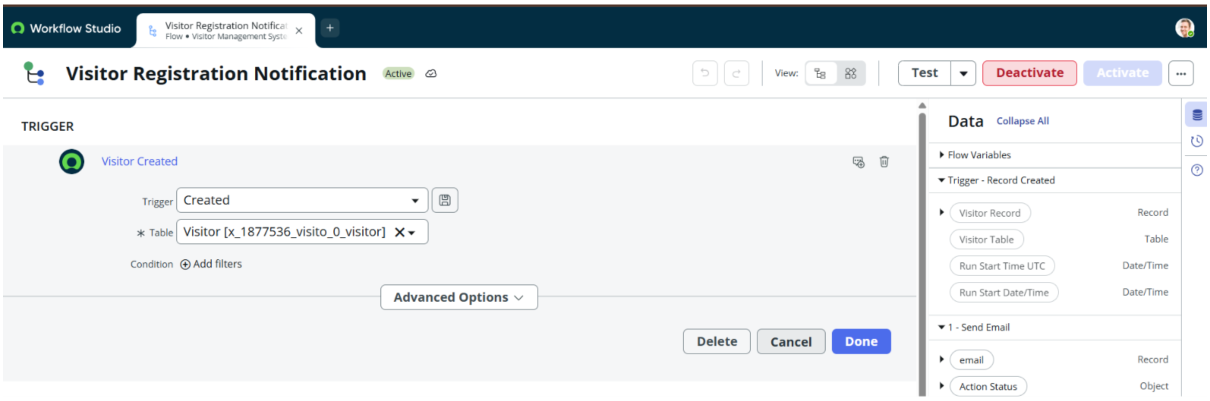
Description

A Flow Designer workflow was created to automatically send an email notification whenever a new visitor record is created in the Visitor Request table.

Steps Performed

1. Opened Flow Designer.
2. Clicked New Flow.
3. Entered the flow name as Visitor Registration Notification.
4. Created a Record Created trigger.
5. Selected the Visitor Request table.
6. Added a Send Email action.
7. Configured the recipient email address.
8. Added visitor details in the email subject and body.
9. Saved and activated the flow.

The flow was successfully configured. Whenever a new visitor is registered, an email notification is automatically sent to the configured recipient with visitor details.

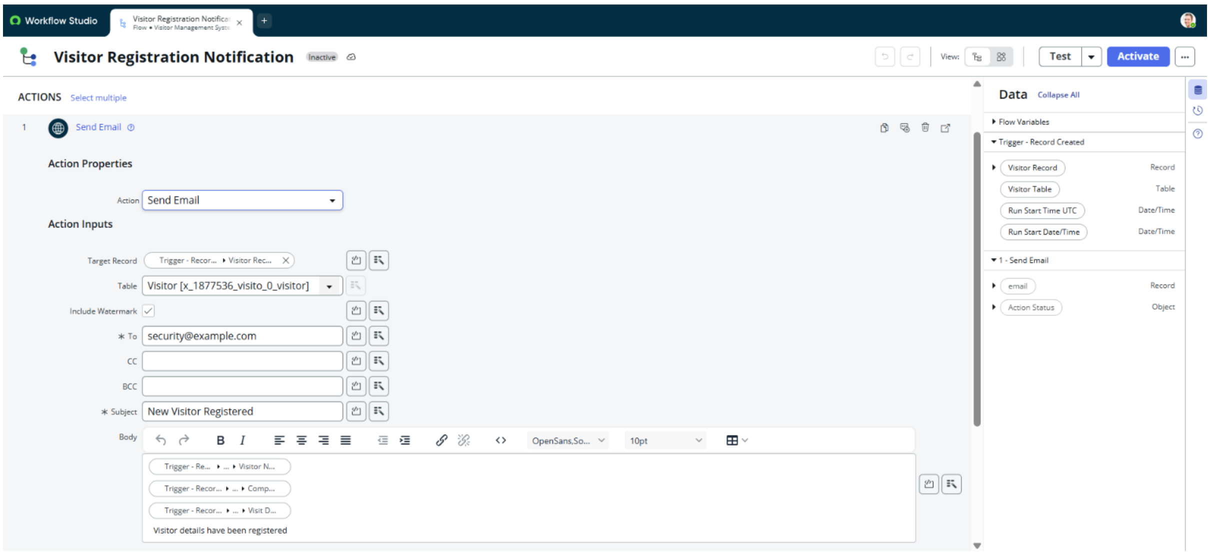


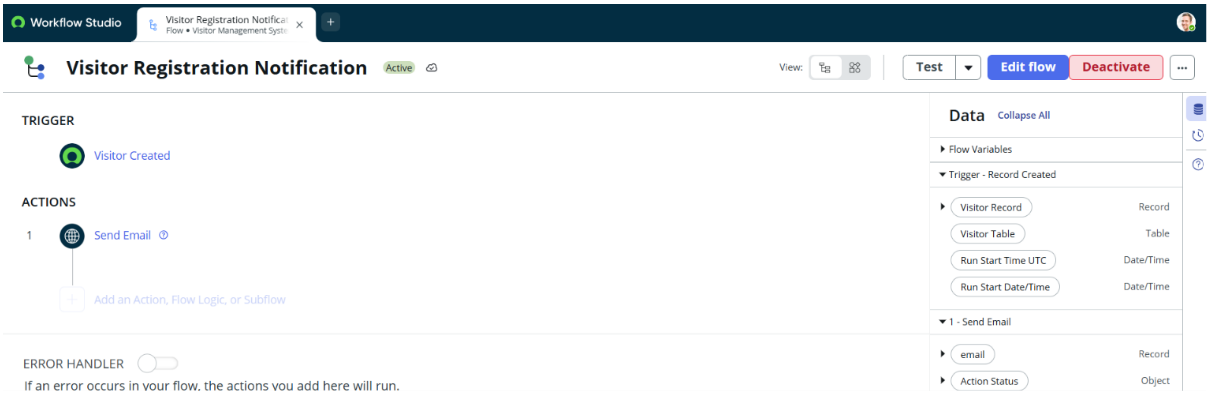
Email Action

Configured Send Email action.

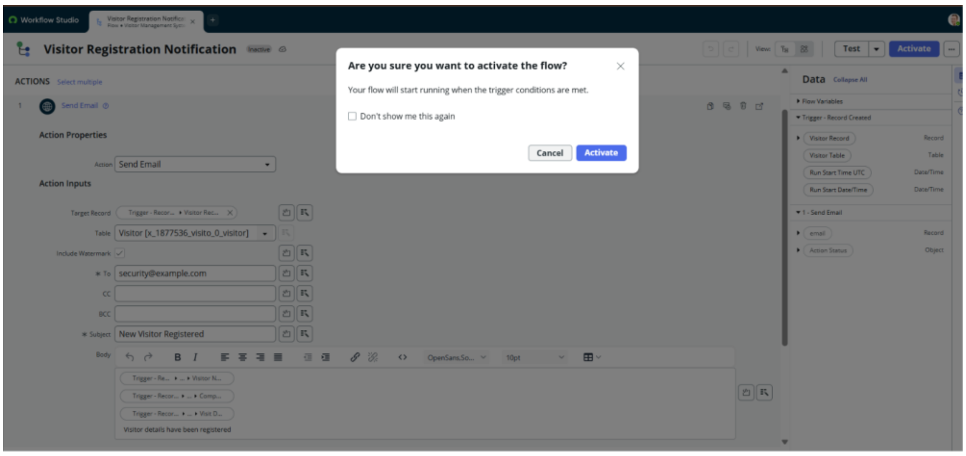
Recipient:

Security Team





Flow Activated:



10. UI ACTION

Step 7: Create Check In Visitor Button

Description

A UI Action named Check In Visitor was created to update the visitor status when the visitor arrives at the organization. This button helps security personnel track visitor check-in activities efficiently.

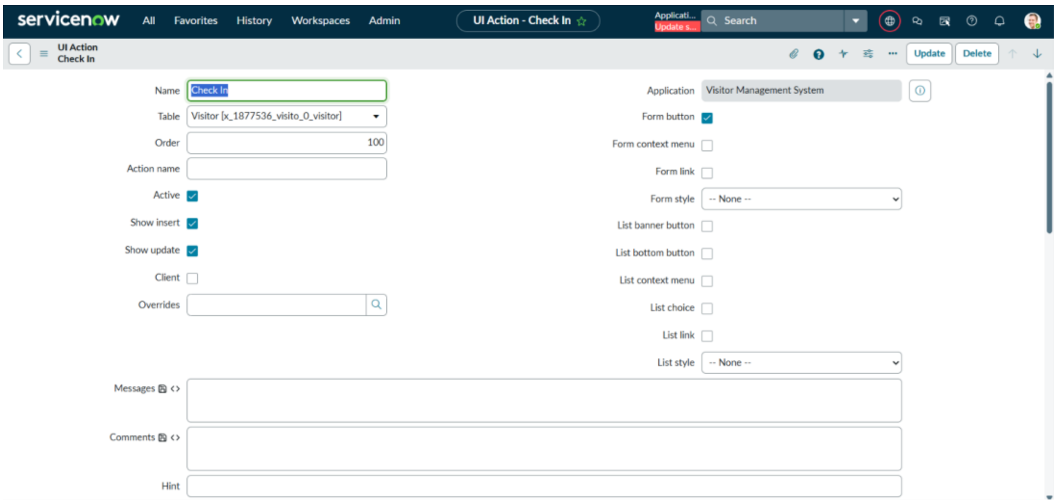
Steps Performed

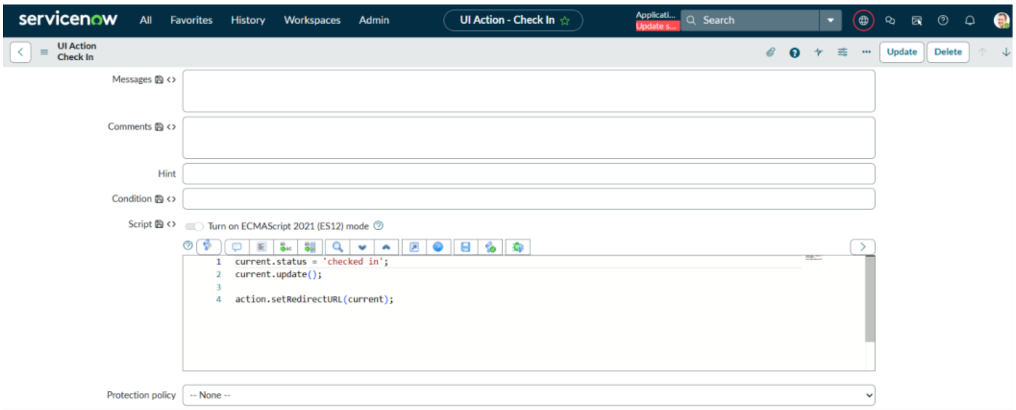
1. Navigated to System Definition → UI Actions.
2. Clicked New.
3. Entered the name as Check In Visitor.
4. Selected the Visitor Request table.
5. Enabled Form Button.
6. Added a script to update the visitor status.
7. Saved and activated the UI Action.

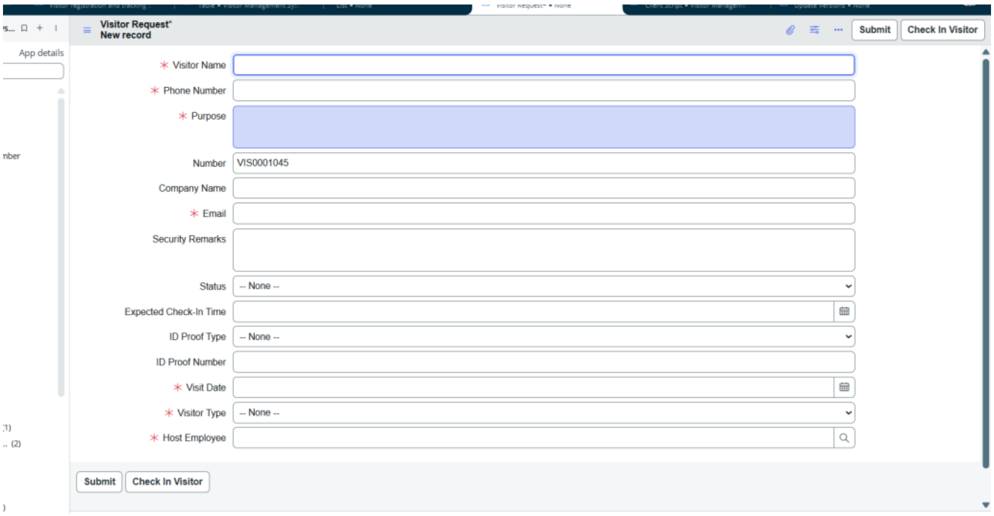
Button Details

Field	Value
Button Name	Check In Visitor
Table	Visitor Request
Type	Form Button
Function	Updates visitor status to Checked In

The Check In Visitor button was successfully created. When clicked, the visitor status is automatically updated to Checked In, allowing accurate tracking of visitor arrivals.







11. REPORTS

Step 8: Create Reports

Reports were created to analyze visitor information and monitor visitor activities. These reports provide useful insights into visitor registrations, visitor status, and visitor categories.

Reports Created

Report 1: Total Visitors

Purpose:
Displays the total number of visitor records created in the system.

Report 2: Pending Visitors

Purpose:
Displays visitors whose status is currently marked as Pending.

Report 3: Checked-In Visitors

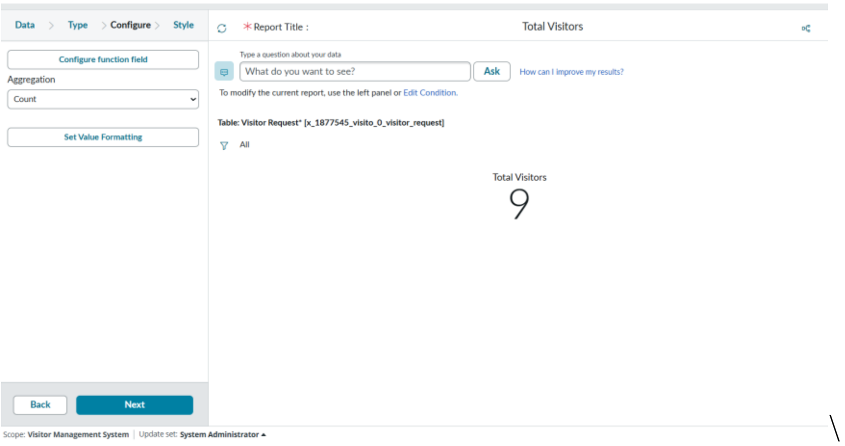
Purpose:
Displays visitors who have successfully checked in.

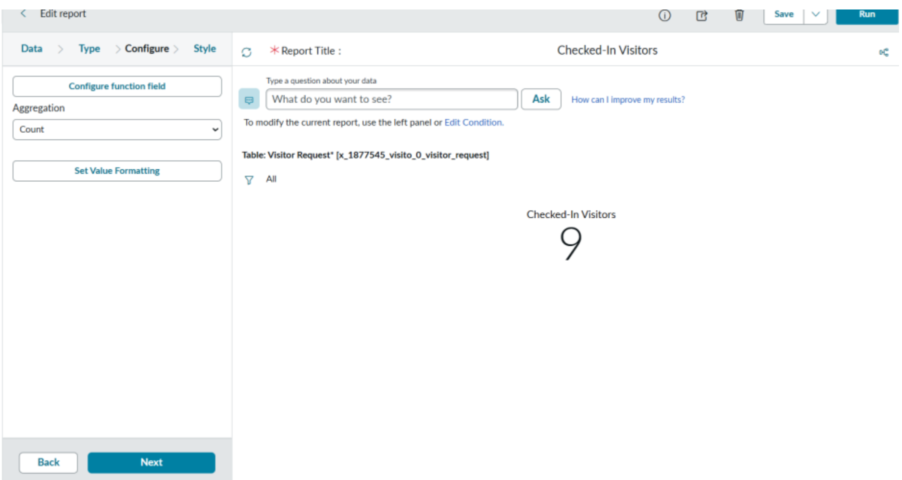
Report 4: Visitors by Type

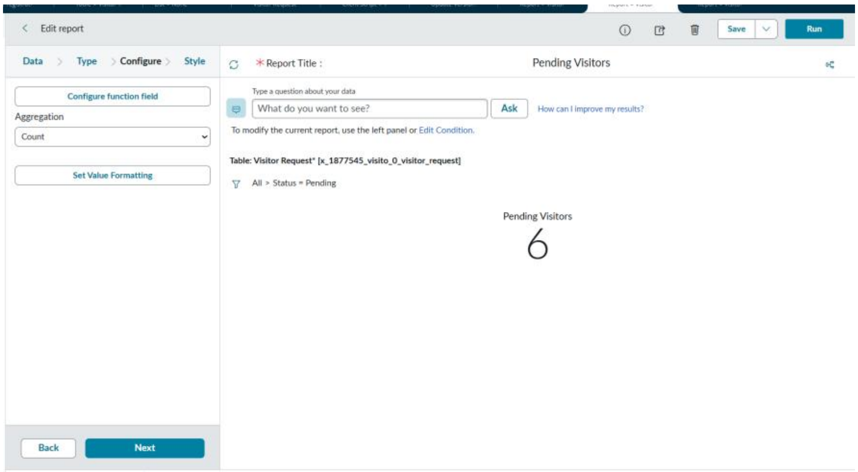
Purpose:
Displays visitor distribution based on visitor type such as Guest, Vendor, Contractor, and Interview Candidate.

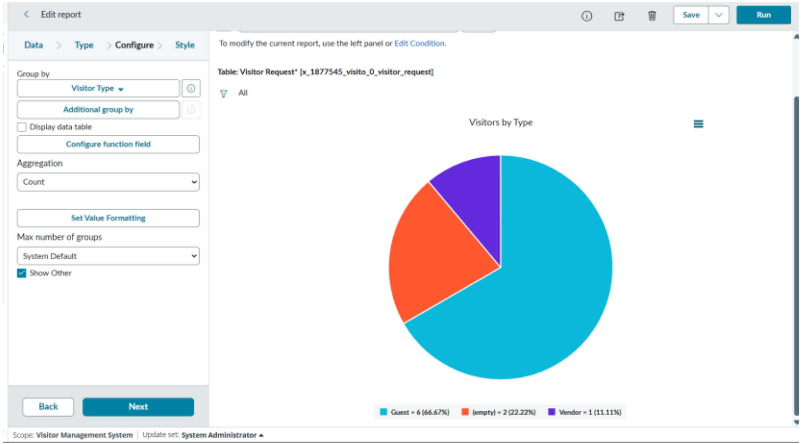
Steps Performed

1. Opened **Reports** in ServiceNow.
2. Clicked **Create New Report**.
3. Selected the **Visitor Request** table.
4. Configured report filters and conditions.
5. Selected the appropriate report type.
6. Saved the report.
7. Repeated the process for all required reports.









12. TESTING

Description

Testing was performed to ensure that all functionalities of the Visitor Management System work correctly.

Test Cases

Test Case	Expected Result	Status
Visitor Registration	Visitor record created successfully	Pass
Phone Number Validation	Invalid phone numbers rejected	Pass
Email Notification	Email notification sent successfully	Pass
Check In Visitor	Visitor status updated correctly	Pass
Reports Generation	Reports displayed successfully	Pass
Dashboard Display	Dashboard loaded successfully	Pass

Outcome

All modules were tested successfully and the application performed as expected.

13. ADVANTAGES

Advantages

- Easy visitor registration process.
- Reduces manual paperwork.
- Improves data accuracy.
- Automated email notifications.
- Real-time visitor tracking.
- User-friendly interface.
- Centralized visitor record management.
- Better security monitoring through visitor logs.

14. CONCLUSION

The Visitor Management System was successfully developed using the ServiceNow Zurich platform. The application automates visitor registration, phone number validation, email notifications, and visitor check-in activities. The system improves efficiency, enhances security, and provides a centralized solution for managing visitor information. The project demonstrates the practical use of ServiceNow application development concepts in a real-world business scenario.

15. FUTURE ENHANCEMENTS

Future Scope

- QR Code based visitor check-in.
- SMS notification integration.
- Visitor ID card generation.
- Mobile application support.
- Biometric verification integration.
- Advanced analytics and reporting.
- Visitor appointment scheduling feature.
- Multi-location visitor management support.